在Raspberry Pi 3上建構簡易Linux系統
Part1:Bootloader & Linux Kernel

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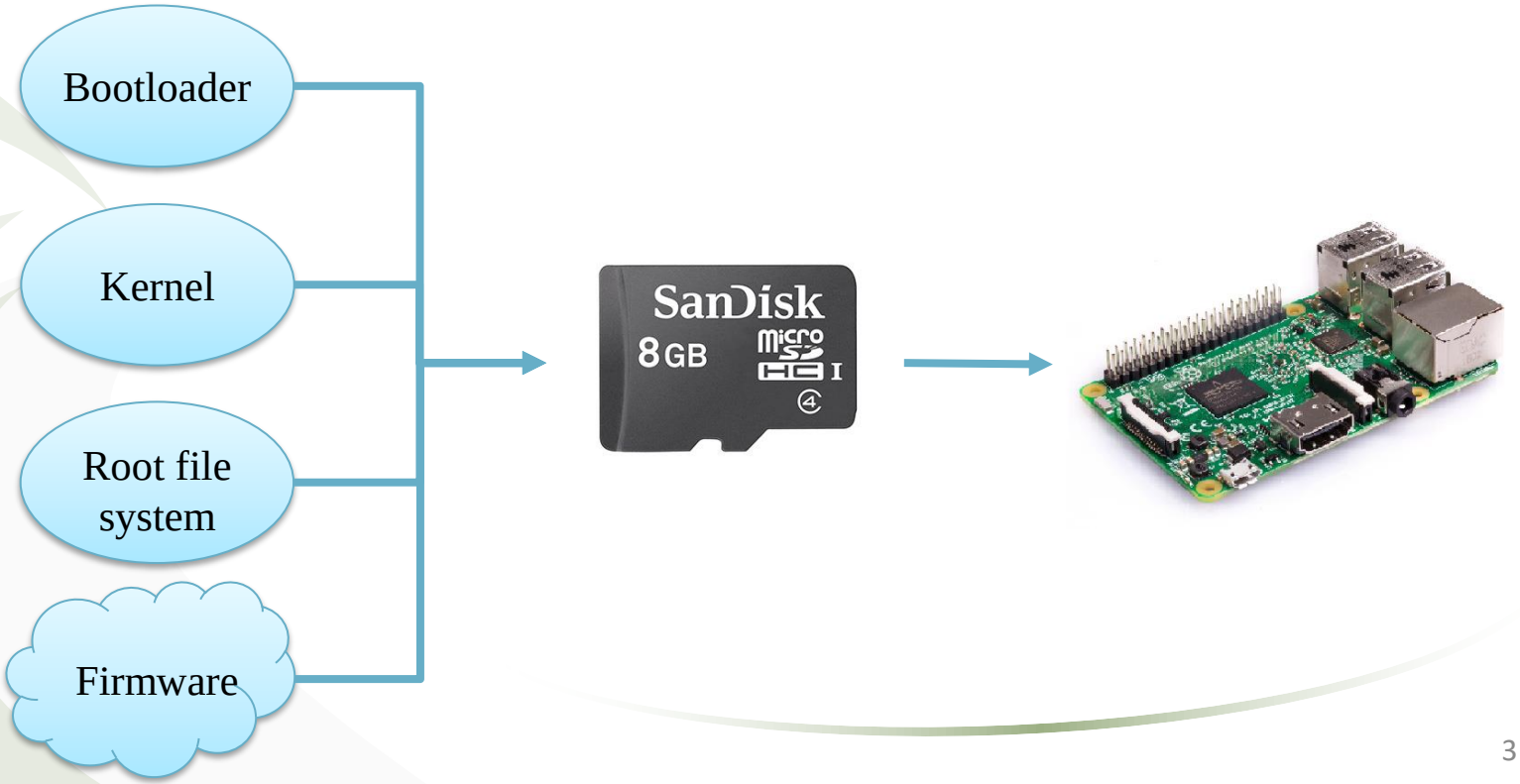
國立中正大學 資訊工程學系

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實驗目的

- 實驗目的：
  - 於Raspberry Pi 3上安裝Linux作業系統。編譯Bootloader、Kernel Image與Root file system，將其存入儲存裝置中，將Raspberry Pi 3開機。



軟硬體與環境需求

- 環境
  - Ubuntu LTS (12.04以上版本)
- Software
  - buildroot
- 硬體
  - Raspberry Pi 3 model B
  - 8G SDcard

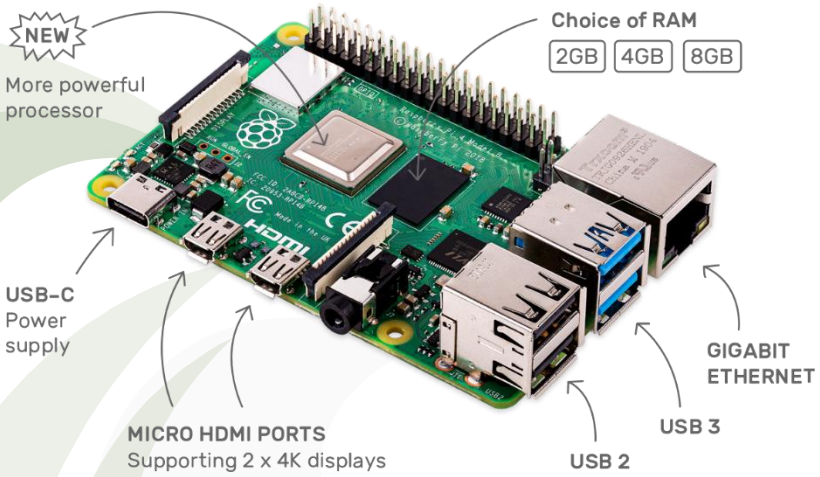
Raspberry Pi 3介紹

- 一款基於Linux的單板電腦，由英國的Raspberry Pi Foundation所開發，以推廣電腦相關之教育學習為主要目標。
- 規格：
  - SoC：Broadcom BCM2837
  - CPU：ARM Cortex-A53 64位元 1.2GHz 四核心處理器
  - 記憶體：1024 MB (LPDDR2)
  - 網路介面：10/100Mbps 乙太網介面（RJ45介面），支援802.11n無線網路及藍牙4.1
  - 電源輸入：5V (通過MicroUSB或經GPIO輸入)





Raspberry Pi 4



- Broadcom BCM2711, Quad core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz
- 2GB, 4GB or 8GB LPDDR4-3200 SDRAM
- 2.4 GHz and 5.0 GHz IEEE 802.11ac wireless, Bluetooth 5.0, BLE
- Gigabit Ethernet
- 2 USB 3.0 ports; 2 USB 2.0 ports.
- Raspberry Pi standard 40 pin GPIO header (fully backwards compatible with previous boards)
- 2 × micro-HDMI ports
- Micro-SD card slot for loading operating system and data storage
- 5V DC via USB-C connector (minimum 3A*)
- 5V DC via GPIO header (minimum 3A*)

Reference from https://www.raspberrypi.com

Raspberry Pi computers and microcontrollers



Raspberry Pi Zero 2 W

Your tiny, tiny $15 computer

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Raspberry Pi 400 Personal Computer Kit

Raspberry Pi 400 is a complete personal computer, built into a compact keyboard.

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Raspberry Pi 400 unit

Raspberry Pi 400 is your complete personal computer, built into a compact keyboard

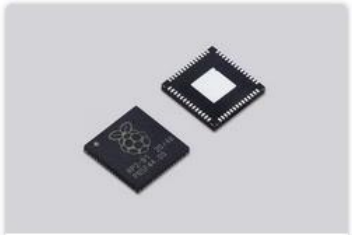
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Raspberry Pi Pico

The new, flexible $4 microcontroller board from Raspberry Pi

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RP2040

A microcontroller chip designed by Raspberry Pi

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Raspberry Pi 4 Desktop Kit

Full desktop computer kit - just connect to HDMI display(s)

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Raspberry Pi 4 Model B

Your tiny, dual-display, desktop computer

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Raspberry Pi 3 Model A+

Our third-generation single-board computer, now in the A+ format

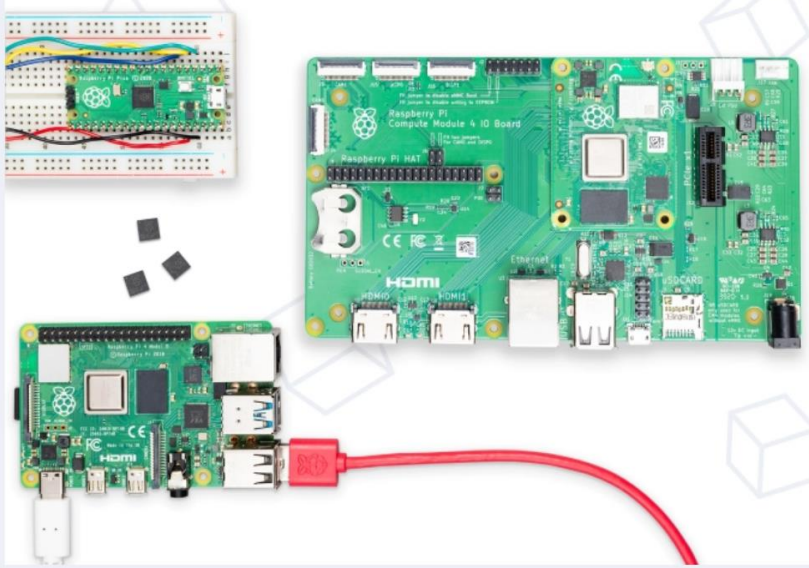
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Reference from https://www.raspberrypi.com

www.raspberrypi.org

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Boot Sequence (1/2)

When the Pi is powered up,

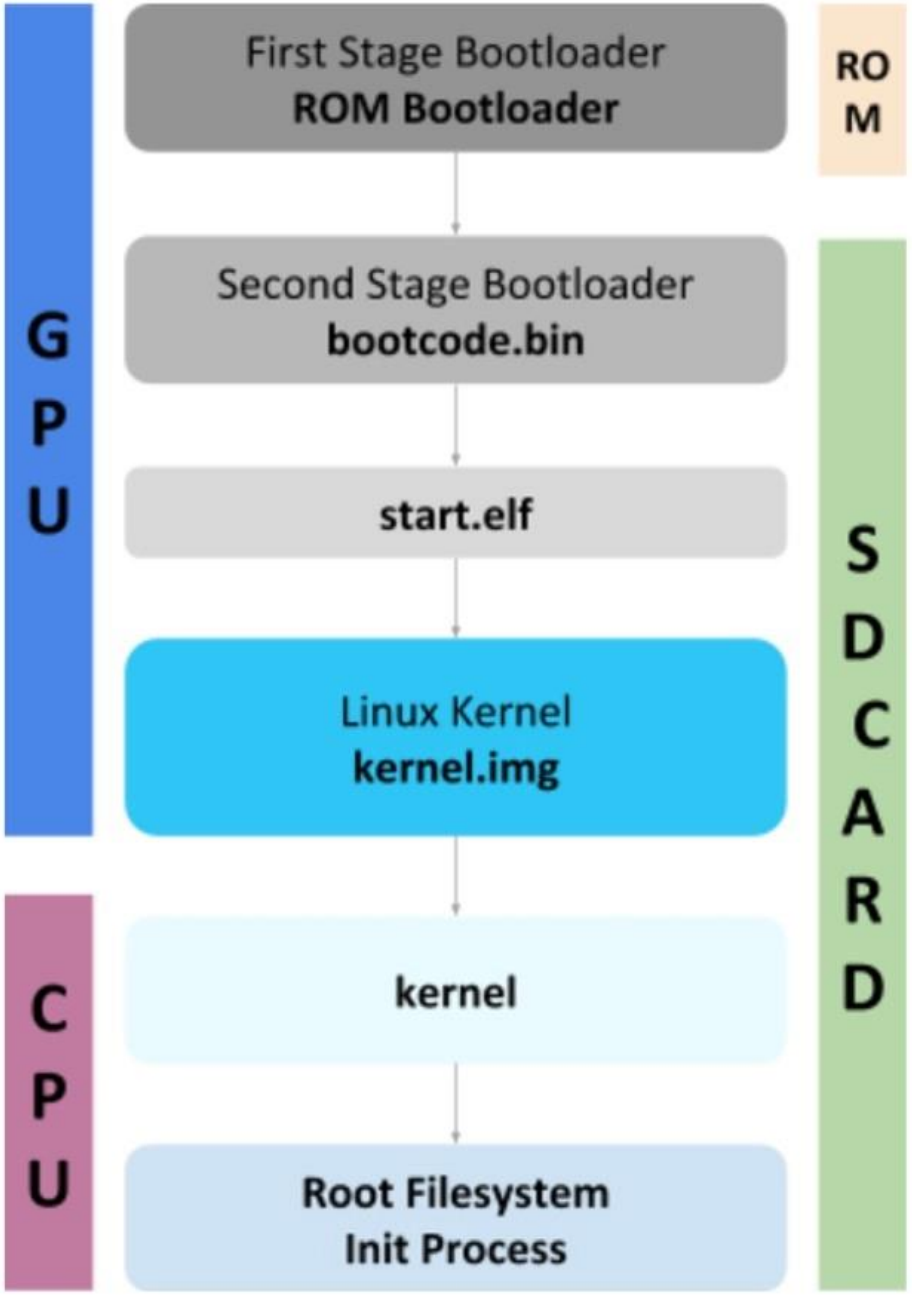
- The ARM processor remains off, SDRAM is disabled, and the GPU core is the one that starts the booting procedure.
- The GPU starts executing the first stage bootloader which is stored in the ROM on the SoC (not on the SD Card).
  - This firmware mainly enables the GPU to access the SD Card, and reads a FAT32 partition on it.

Reference from Ahmed El-Arabawy's slide

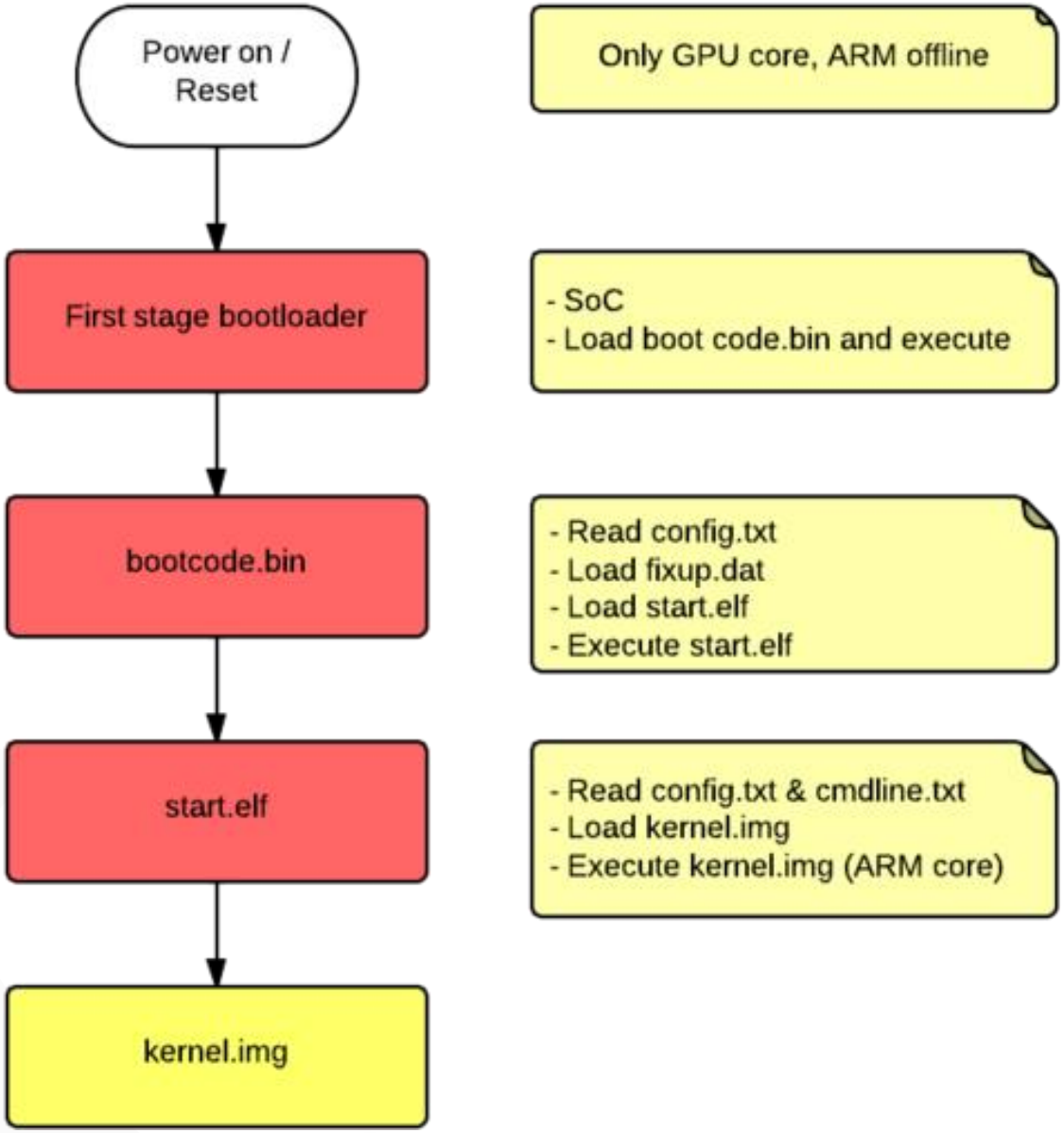
Boot Sequence (2/2)

- The first file read from the SD Card is the second stage bootloader (bootcode.bin). The GPU reads this file and executes it.
- The job of this stage of the bootloader is to enable the SDRAM and it loads and runs start.elf.
- At the last bootloader stage, start.elf reads the kernel image (kernel.img), configuration file(config.txt), and kernel command line parameters (cmdline.txt), then it loads them in memory, and wakes up the ARM core.

Reference from Ahmed El-Arabawy's slide

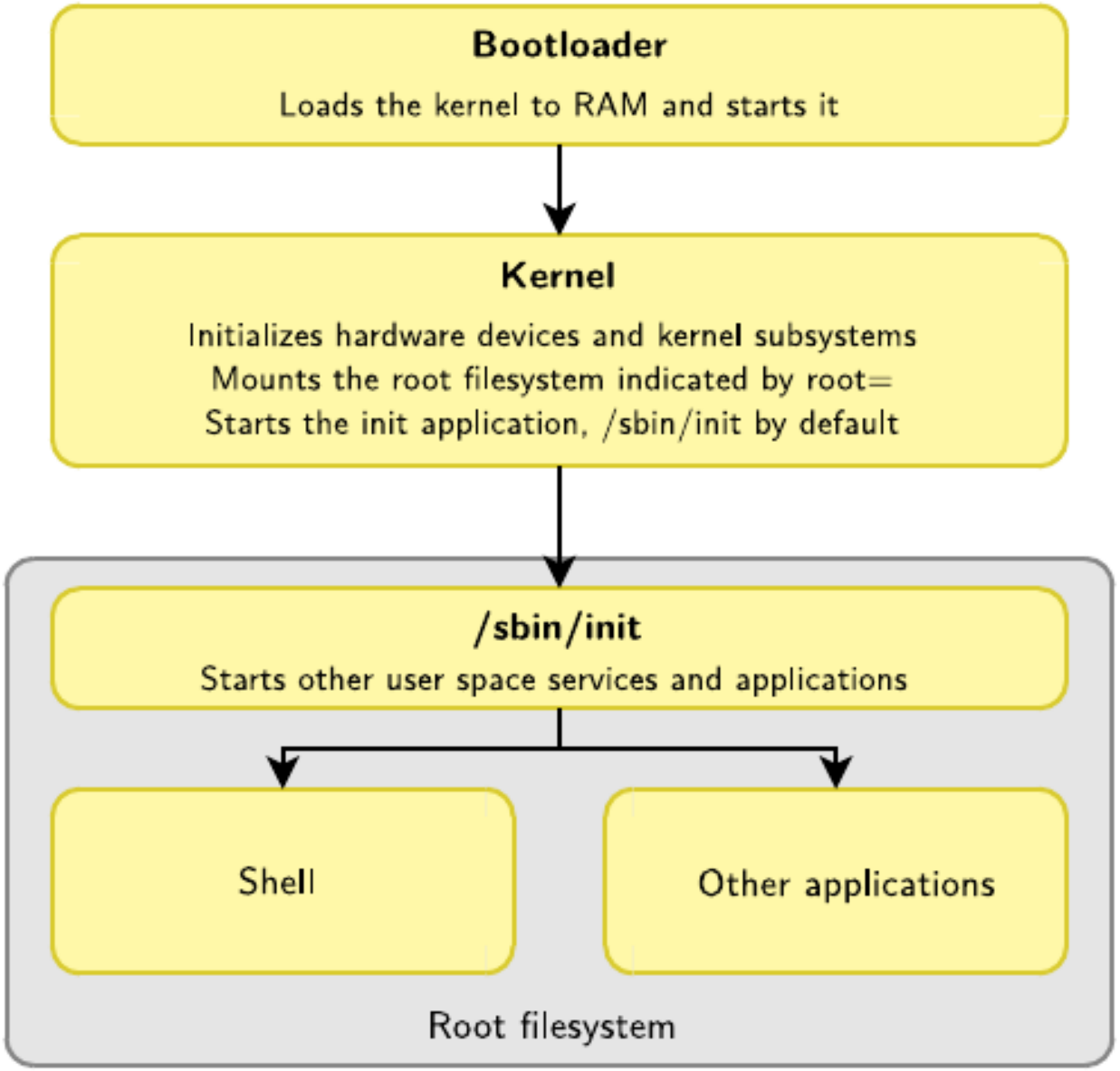


Reference from https://argus-sec.com/raspberry-pi-remote-flashing/



Reference from https://github.com/gogojesse/gos/wiki/Raspberry-Pi

簡易Linux系統架構



Raspberry Pi Partitions

BOOT	RASPBERRY (rootfs)		
FAT32	EXT4		
• bootcode.bin • start.elf • Image • fixup.dat • config.txt • cmdline.txt • *.dtb	• /bin • /boot • /dev • /etc • /home • /lib	• /media • /mnt • /opt • /proc • /root • /run	• /sbin • /srv • /sys • /tmp • /usr • /var

Reference from https://argus-sec.com/raspberry-pi-remote-flashing/

實驗流程

- 使用Buildroot編譯出：
  - Bootloader
  - Kernel
  - Root file system
- 切割並格式化SD card，將Bootloader、Kernel Image與Root file system 放入儲存裝置中
- 啟動與測試

Buildroot介紹

- Target：
  - Making Embedded Linux Easy
- 基本概念：
  - 建立一個自用的cross-compilation toolchain， 再經由此cross-compiler編譯Kernel與Root file system
- 主要設計理念：
  - Simple to use
  - Simple to customize
  - Reproducible builds
  - Small root file system
  - Relatively fast boot
  - Easy to understand
- 官網：https://buildroot.org/



Buildroot

- Buildroot doesn't include runtime package management - it's designed for building appliances really - build a fixed, known working system from a configuration file. So if you're wanting to build an OS where security patching is just a few commands away, Buildroot is not for you.
  - No APT.
  - If you need “apt”, try Raspberry Pi OS (Raspbian).

- If you want to build a lightweight OS for the Raspberry Pi that performs a specific function, then Buildroot could well be for you

實驗流程

- 下載buildroot
  - **$ git clone https://gitlab.com/buildroot.org/buildroot.git**

```
pschen@debian10x:~/x$ git clone git://git.busybox.net/buildroot
Cloning into 'buildroot'...
remote: Enumerating objects: 13607, done.
remote: Counting objects: 100% (13607/13607), done.
remote: Compressing objects: 100% (9618/9618), done.
remote: Total 458173 (delta 7568), reused 7351 (delta 3945), pack-reused 444566
Receiving objects: 100% (458173/458173), 99.64 MiB | 446.00 KiB/s, done.
Resolving deltas: 100% (320296/320296), done.
Checking out files: 100% (12899/12899), done.
```

- Configure buildroot，使其能夠生成可直接使用的：
  - Bootloader
  - Linux kernel

Bootloader & Kernel

- 安裝執行 menuconfig 所需要的libncurses

```
$ sudo apt install libncurses-dev
```

- 調整Buildroot設定

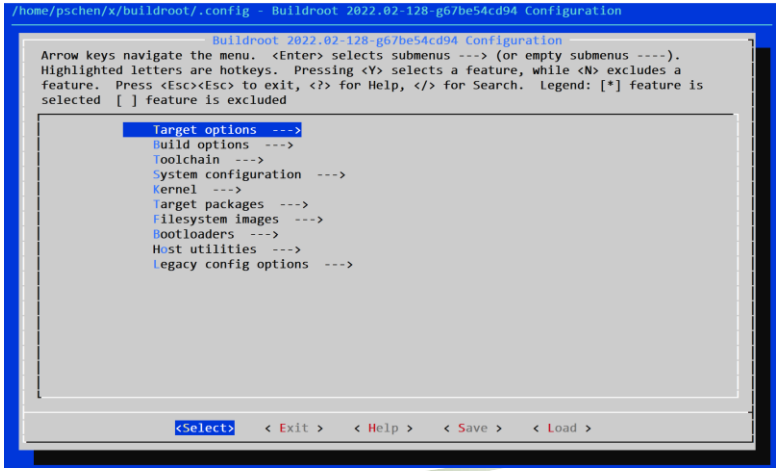
```
$ make menuconfig
```

或

```
$ make raspberrypi3_64_defconfig
```

- 調整Linux設定

```
$ make linux-menuconfig
```



Bootloader & Kernel

- **Buildroot設定:**可直接使用Buildroot自帶的、適用於Raspberry Pi3的預設定值，位於**configs/raspberrypi3_64_defconfig**。內容僅包含與預設值不同的部分，內容相當簡潔，可以直接觀察。

```
$ make raspberrypi3_64_defconfig
```

- **Linux設定:**使用預設值即可運行，日後有特定需求可自行調整(例如: 把kernel精簡化，移除不必要的模組)。

References

- https://www.raspberrypi.com/documentation/computers/getting-started.html
- http://fichugh.blogspot.tw/2016/02/buildroot-study.html
- http://www.embeddedforu.com/embedded-linux/raspberry-pi/embedded-linux-development-on-raspberry-pi-using-buildroot-part1/
- http://www.embeddedforu.com/embedded-linux/raspberry-pi/embedded-linux-development-on-raspberry-pi-using-buildroot-part2-2/
- https://github.com/BrianSidebotham/rpi-buildroot
- https://buildroot.org/